🧪 Chemical Bonding Overview

1. Why Atoms Bond

- Atoms bond to achieve a **stable electron configuration**, often following the **octet rule** (8 electrons in the outer shell).
- Bonding lowers the overall energy of the system, making compounds more stable than isolated atoms.

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2. Types of Chemical Bonds

Bond Type	How It Forms	Key Features	Examples
Ionic Bond	Transfer of electrons from one atom to another	Produces oppositely charged ions held by electrostatic attraction	NaCl (sodium chloride)
Covalent Bond	Sharing of electron pairs between atoms	Strong, directional bonds; can be single, double, or triple	H_2O, CO_2
Metallic Bond	Delocalized "sea of electrons" shared among metal atoms	Explains conductivity, malleability, and luster of metals	Cu, Fe

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3. Bond Properties

- **Bond Energy**: Energy required to break a bond; higher bond energy = stronger bond.
- **Bond Length**: Distance between nuclei of bonded atoms; shorter bonds are stronger.
- **Electronegativity**: Determines bond polarity; large differences favor ionic bonds, small differences favor covalent bonds.

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4. Special Concepts

- **Polar Covalent Bonds**: Unequal sharing of electrons (e.g., H–O in water).
- **Nonpolar Covalent Bonds**: Equal sharing of electrons (e.g., H–H).

- **Exceptions to Octet Rule**: Some atoms (like H, B, and transition metals) don't follow the octet strictly.
- **Resonance Structures**: Some molecules (like benzene) are best represented by multiple Lewis structures.

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5. Importance of Chemical Bonding

- Explains **molecular structure** and **properties of matter**.
- Determines **reactivity** and **stability** of compounds.
- Forms the basis of **organic chemistry, materials science, and biochemistry**.

📘 Summary

Chemical bonding is the glue of matter, allowing atoms to form molecules and solids. Ionic bonds transfer electrons, covalent bonds share them, and metallic bonds create a communal electron "sea." Understanding bonding is essential for predicting chemical behavior, designing new materials, and mastering the periodic trends of elements.